

CSDA 5110
Fall 1 2024
Homework 2

Read chapters 5 - 8 from the textbook.

Complete the homework using R Studio. You will need to upload the Rmarkdown file containing the code you create.

It **MUST** be the Rmarkdown file. The file you create will have a ".rmd" extension.

You should knit the markdown file as a word file. It should contain all the output.

You can edit this file to put answers. Save as pdf and upload it.

You are uploading .rmd file and a pdf file with complete answers.

1. **Create a new R markdown. Using comments, write your name, the date, and “CSDA 5110 Homework 2” at the top of the script. Write your answers to the rest of these exercises in this script and be sure to copy and paste the original questions using comments! Your script should only contain valid R code and comments.**
2. Which (if any) of the following variable’s names is/are invalid?

```
thisone <- 1  
THISONE <- 2  
1This <- 3  
this.one <- 4  
This.1 <- 5  
ThIS.....ON...E <- 6  
This!On!e <- 7  
lkjasdfkjsdf <- 8
```

3. 2021 was a good year for the price of Gold – your investment portfolio made \$100,800 from investing in gold. Create a variable called **gold.in.2021** and assign the value of your investment portfolio to it.
4. During the last bank audit, we discovered that there was an error caused by a system calculation error and \$5000 per year wasn’t added to your gold investment portfolio for the years 2018, 2019, and 2020. Create a new variable called **calcError** and assign it the amount miscalculated by the system. Create another variable called **corrected.total** and add the variables **gold.in.2021** and **calcError** to this new variable. What is the new total amount?
5. Create a variable called numberVector1 and numberVector2. Using the **c()**, assign a vector of numbers, 1 through 15 to each vector. Use a different method of creating the vector for each variable. **(Hint: this was covered in your reading of chapter 4)** Use the appropriate function to check the length of each of the vector variables you created above
6. Using the two vector variables created above, add them together and assign the total to a variable called totalVectors.
7. Use the appropriate function to sum the totalVectors variable to display the sum of the totalVectors variable.

8. Joanne thinks that she completes more tasks when she's had a cup of coffee than when she doesn't. To test this, she recorded how many tasks she completed over 7 days without drinking any coffee, and then did the same over 7 days while drinking coffee. Her results are stored in a comma delimited CSV file called Joannes Tasks
 - a. Import the comma delimited CSV file into R and assign the data to a variable called joanneTasks
 - b. Verify the variable holding the data is a data.frame variable
 - c. Using the appropriate function, write the code to verify that there are 7 rows and 4 columns in the data frame.
9. Use the appropriate function that will display summary descriptive statistics about each column within the data frame.
 - a. In your code, create comments about information you deduced from the summary descriptive statistics.
10. Using the appropriate syntax write the code that allows you to access the column that contains the Number of Tasks Completed Without Coffee data.
 - a. Use the appropriate function to display the mean value of that column.
 - b. Use the appropriate function to display the total of the values in the column.
11. Using the appropriate syntax write the code that allows you to access the column that contains the Number of Tasks Completed With Coffee data.
 - a. Use the appropriate function to display the mean value of that column.
 - b. Use the appropriate function to display the total of the values in the column.
12. Using the appropriate syntax write the code that allows you to access the column that contains the Total Daily Tasks data.
 - a. Use the appropriate function to display the mean value of that column.
 - b. Use the appropriate function to display the total of the values in the column.
13. Write a function that will display the question below and return the correct answer.
 - a. Was Joanne able to complete more tasks when she drank coffee or days when she did not drink coffee?

Use Florida House Sale Data for the remaining questions.

14. Select **two** quantitative variables (from Taxes, Price, Size, and Lot)
 - a. Draw histogram for the two selected quantitative variables.
 - b. Draw the density plot for the two selected quantitative variables.
15. Select the categorical variable Quadrant.
 - a. Draw Pareto chart.
16. For **both the quantitative** variables used in question 14.
 - a. Determine the mean and median.
 - b. Compute the standard deviation of both variables.
 - c. Compute five number summary for both the variables.
 - d. Draw boxplots.